

Web Server Enumeration and Vulnerability Assessment

Project Goal

The aim of this project is to get familiar with Linux, Networking, Nmap and fundamentals of Security. More specifically, we will be able to answer questions about web servers like:

- a) Which ports are open?
- b) Which web server is running?
- c) Which version?
- d) Which HTTP methods are enabled?
- e) Are dangerous methods enabled?
- f) Is directory listing enabled?
- g) Is HTTPS configured?
- h) What information leaks through HTTP headers?

Phase 1 – Understanding the Environment

Before heading deep into our project, we must first understand the general principles by answering simple but significant questions.

What is Apache?

- Apache is an application that listens to HTTP requests from clients (browsers) and responds by serving web pages, files or dynamic content.

When a client (browser) visits `http://192.168.1.153` which port will the browser connect to?

- Since it is an HTTP service, the port the browser connects to will be port 80.

Phase 2 – Apache Setup

Before starting the enumeration, we need a web service on our Ubuntu VM. So, we install Apache2 with the command **sudo apt install Apache2**, after checking for any Ubuntu updates available using **sudo apt update**.

Phase 3 – Testing

After installing Apache2 on our VM, we were ready to proceed to the testing phase. First with the command **nmap -sT -Pn 192.168.1.153** we were able to see all the TCP connections on the destination IP. Having verified that our HTTP service is running, we also saw that it used the commonly known HTTP services' port, **port 80**. Then, to start the enumeration we used the **nmap -sT -Pn --script "http-*"** command, which uses all the HTTP-related NSE scripts. However, our enumeration does not require all the HTTP-related scripts, just the ones about the **HTTP-methods**, the service's **version**, **any interesting directories available from it**, the **page title** and the **headers**.

Phase 4 – Results and Conclusions

The commands and their results:

nmap -sT -Pn --script "http-methods" 192.168.1.153:

- http-title: Apache2 Ubuntu Default Page: It works

nmap -sT -Pn --script "http-headers" 192.168.1.153:

- | Date: Wed, 01 Jul 2026 07:48:20 GMT | Server: Apache/2.4.66 (Ubuntu) | Last-Modified: Wed, 01 Jul 2026 06:53:31 GMT | ETag: "29b0-6558726475499" | Accept-Ranges: bytes | Content-Length: 10672 | Vary: Accept-Encoding | Connection: close | Content-Type: text/html | |_ (Request type: HEAD)

nmap -sT -Pn --script "http-methods" 192.168.1.153:

- | Supported Methods: GET POST OPTIONS HEAD

nmap -sT -Pn --script "http-enum" 192.168.1.153:

- There are no common directories or files

Conclusion:

Since the methods of HTTP do not involve **PUT and/or DELETE**, they are safe. Also, the page title, the directories and the headers do not show any sensitive information. Overall, the risk of attack based on the evidence is low.